Module 10

Applying Network Security Features

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Learning Objectives

- Explain identity and access management concepts.
- Distinguish protocols and standards used for authentication and directory management.
- Use defense in depth techniques to ensure that only policy-compliant devices can connect to the network.
- Apply security rules, such as ACLs and content filtering, to manage network traffic.

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Lesson 10.1

Authentication

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Discussion

Boarding Pass

Passenger Name: Samantha Simons
Birthdate: 1-11-2001

Required to board:

- Passport
- Printed Ticket

Destination: Aruba
Ship: The Splash
Deck: 12
Cabin: 12345

Included:

✓ Beverages: All-inclusive
✓ Meals: A la carte
✓ Scuba diving excursion
✓ Resort shopping excursion

All onboard activities and purchases will be logged in your customer account.

Identity
Who she claims to be

Authentication
Proof that she is who she claims to be

Authorization
Where she is allowed to go and what level of access she will have once onboard

Accounting
Method for tracking and logging activity

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Think About It: Access Control

Identity	An account or ID that uniquely represents a user, device, or process on the network
Authentication	Factor(s) used to prove a subject is who or what it claims to be
Authorization	Rights and permissions a subject is granted within a system or network
Accounting	Tracking of authorized/unauthorized usage of a resource by a subject

What are some examples of each of these in network security?

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Authentication Methods

Something you know

Password

Something you have

Smart card

Something you are

Fingerprint

Something you do

Gait

Somewhere you are

Geofencing

Somewhen you are

Time restrictions

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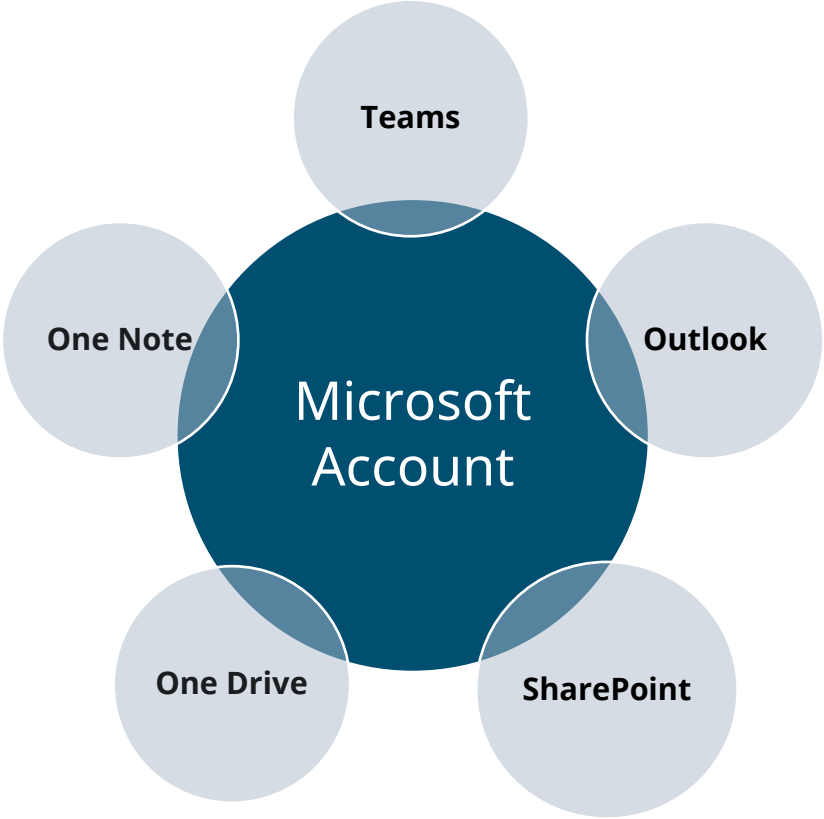
Single Sign-On



I'm so happy I don't have to log in to all these apps one at a time!

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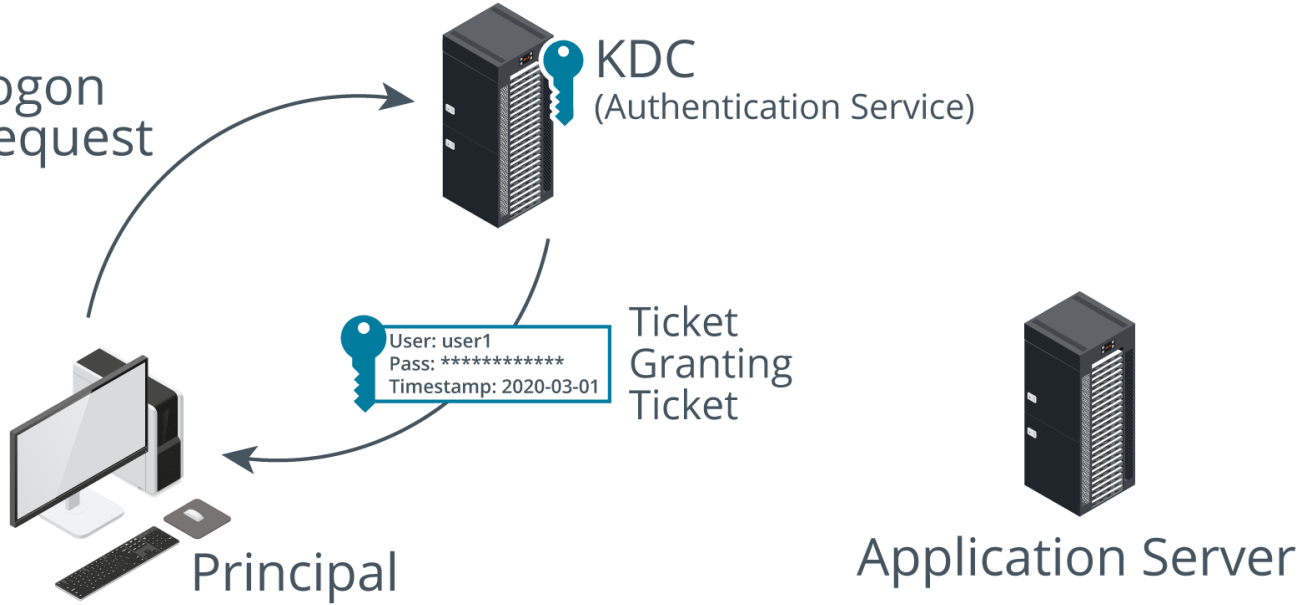
Kerberos SSO Authentication



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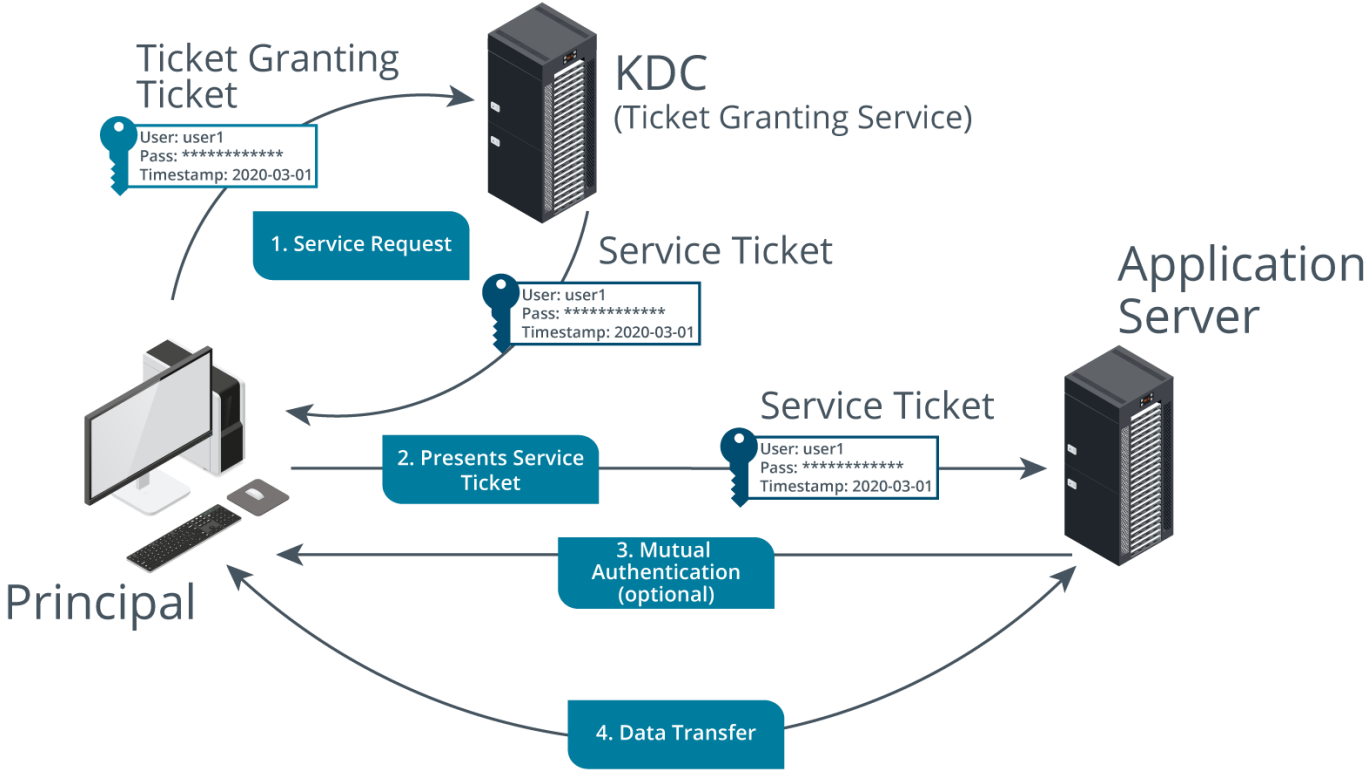
Kerberos Ticket Granting Service



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Lesson 10.2

Authentication and Account Management

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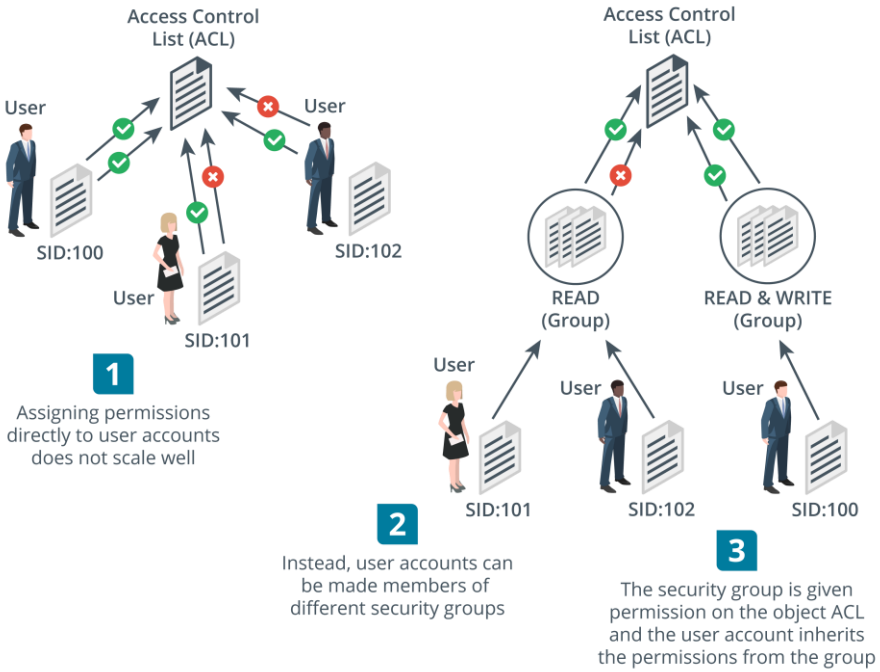
Access Control Models

Discretionary Access Control (DAC)

Owner-controlled access



Role-Based Access Control (RBAC)

Role-based access

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Access Management: Terms to Know

Privileged account

- Account for major system changes

Privileged Access Management (PAM)

- Controls to prevent abuse of privileged accounts

Least Privilege

- Granting users just enough rights to perform their job

Separation of Duties

- Dividing tasks to prevent misuse

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Think About It: Who Needs Access?

System Administrator	Developer	HR Manager	Sales Representative

Employee records	HR software
Financial database	Accounting software
Server configurations	Development tools
Office 365	Customer database
Code repository	Sales software
Security settings	Payroll software

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Lesson 10.3

Network Hardening

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Defense in Depth Model

Data	Access Controls Lists
Application	Application hardening, antivirus
Host	OS hardening, patching, BIOS password
Internal Network	Subnets, VLANs, IDS/IPS
Perimeter	Firewalls, VPN, IDS/IPS
Physical Security	Guards, locks, tracking devices, fences
Policies, Procedures, and Awareness	User education, security documents

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Device and Service Hardening

Each host and network device should be hardened.

What are some ways you can harden devices?

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Switch Security

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Switch Security – Terms to Know

Network access control (NAC)

- System to authenticate/authorize access to a network

Switch

- Connects/manages communication on a network

Switch Ports

- Dedicated interface on a switch that connects devices

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Port Security

- Secure physical access
- Regularly update switch firmware
- Disable unneeded ports
- Configure MAC filtering
- Enable Router Advertisement Guard (RA)
- Configure DHCP snooping
- Implement 802.1x authentication (EAP)

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Think About It: What Would You Do?

ALERT! An unauthorized device was found connected to the network in the server room.

To prevent such occurrences, you decide to secure _____ access, using a secure server/equipment room and/or lockable cabinets).

You disable _____ to prevent unauthorized devices from being connected to the switch.

Additionally, you plan to enable _____ to limit the connections to known devices.

Finally, you enable _____ to inspect DHCP traffic for authenticity.

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Lesson 10.5

Network Security Rules

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Firewalls and Proxy Servers

Firewalls	Proxy Servers
• Filters all traffic and blocks/allows based on ACL • Ensures only permitted traffic • Filters traffic entering/leaving a network segment	• Deconstructs/analyzes packets to ensure rule compliance • Enforces company policies • Manages/redirects requests to control internet access

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Content Filtering

Firewall Filtering Criteria

- Protocol
- Source and destination address
- Source and destination port
- Stateful inspection

Proxy Server Filtering Criteria

- URLs and domain names
- Content keywords
- Bandwidth usage
- Reputation of website

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Common Firewall Misconfigurations

- Overly permissive ACL
- Overly prohibitive ACL
- Failure to apply an implicit deny
- Improper logging settings

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Think About It: Two Truths and a Lie

Firewalls deconstruct and analyze packets to ensure rule compliance before forwarding.

Firewalls are positioned to filter traffic entering or leaving a network segment.

Proxy servers can filter based on content keywords.

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Summary and Review

- **Access control**: Only authorized users and devices can access resources (physical & digital measures).
- **Defense in depth**: Layers of security (policies & mechanisms) safeguard data integrity, confidentiality, and availability.
- **Firewalls & Proxies**: Filter traffic and manage requests for network security and performance.
- **Content Filtering**: Fine-tune traffic with rules based on protocols, addresses, content, and more.

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